

Plan of Attack: Section 1, Raw Data Description

Costco Australia / Synthetic Control

Research question: Does the entry of a Costco gas station into an Australian local market cause nearby competitor stations to lower their retail fuel prices?

1. Raw Data Description

We use four datasets, described separately below: three publicly accessible Australian state fuel-price registries that we combine into a unified station-month panel, plus a hand-collected list of Costco fuel-station opening dates. The three registries together yield approximately 17.4 million station-day price records across 6,084 unique stations between January 2018 and April 2026.

1.1 NSW FuelCheck

NSW FuelCheck is maintained by the New South Wales Government and has required mandatory price reporting from every retail fuel station in NSW (and, since November 2023, the ACT) since August 2016. Historical records are published as monthly Excel files via data.nsw.gov.au; we pulled 93 monthly files covering December 2016 through January 2026, totalling **4,401,971 price-update records** with 8 columns. Each row is one price-update event: a station $\times$ fuel type $\times$ the timestamp at which the operator changed (or re-confirmed) its posted price. FuelCheck stores price-change events rather than daily snapshots. Composite unique key: (ServiceStationName, Address, FuelCode, PriceUpdatedDate). The file contains 4,813 unique stations (4,543 after the name normalization used in the live-API coordinate join, see Section 2(f)), 62 unique brands, 565 unique postcodes, and 11 unique fuel-type codes (regular unleaded ULP/U91/E10, premium PUL95/PUL98, diesel DSL/PDL, LPG, and others). Coordinates are not in the historical files; we attached lat/lng by joining station name + postcode to a one-time pull from the live FuelCheck API.

Field	Description
ServiceStationName	Station name (operator-provided)
Address	Street address
Suburb	Suburb
Postcode	Postcode (4-digit)
Brand	Brand affiliation (Costco, Caltex, Shell, BP, etc.)
FuelCode	Fuel-type code (U91, E10, PUL95, PUL98, DSL, LPG, ...)
PriceUpdatedDate	Timestamp at which price was reported
Price	Posted retail price (cents per liter)

1.2 Queensland Fuel Price Reporting Scheme

The Queensland Fuel Price Reporting Scheme is administered by Queensland Treasury; mandatory station reporting began in December 2018. Historical records are published as monthly CSVs via data.qld.gov.au; we pulled 85 monthly files covering December 2018 through December 2025, totalling

4,970,650 price-change records with 13 columns. Each row is one price-change event: a station × fuel type × the UTC timestamp at which the price changed. Composite unique key: (`SiteId`, `Fuel_Type`, `TransactionDateutc`). The file contains 1,883 unique stations, 37 brands, 375 postcodes, and 10 fuel-type strings. Coordinates are present inline.

Field	Description
<code>SiteId</code>	Stable per-station numeric identifier
<code>Site_Name</code>	Station name
<code>Site_Brand</code>	Brand affiliation
<code>Sites_Address_Line_1</code>	Street address
<code>Site_Suburb</code>	Suburb
<code>Site_Post_Code</code>	Postcode (4-digit)
<code>Site_Latitude / Site_Longitude</code>	WGS84 coordinates (used for haversine distance to Costco)
<code>Fuel_Type</code>	Fuel-type label (Unleaded 91, e10, Premium 95, ...)
<code>Price</code>	Posted retail price in tenths of a cent (1899 = 189.9 ¢/L)
<code>TransactionDateutc</code>	UTC timestamp of the price change

1.3 WA FuelWatch

WA FuelWatch is maintained by the Western Australia Government and has required daily mandatory price reporting from every retail fuel station in WA since January 2001, the longest fuel-price history of any Australian state. Historical records are published as monthly CSVs via data.wa.gov.au; we pulled 100 monthly files covering January 2018 through April 2026, totalling **8,047,153 daily-snapshot records** with 11 columns. (We chose not to pull the full 2001–2017 archive because no usable Costco event has a pre-period that benefits from extending earlier than 2018.) Each row is a daily snapshot (a station × fuel type × calendar day), and unlike NSW and QLD, FuelWatch publishes one record per station per day regardless of whether the price changed. Composite unique key: (`TRADING_NAME`, `ADDRESS`, `PRODUCT_DESCRIPTION`, `PUBLISH_DATE`). The file contains 1,423 unique stations, 47 brands, 200 postcodes, and 7 fuel-type strings. Coordinates are not in the historical files; we attached lat/lng by joining station name + postcode to a one-time pull from the live FuelWatch API.

Field	Description
<code>PUBLISH_DATE</code>	Calendar day of the snapshot
<code>TRADING_NAME</code>	Station name
<code>BRAND_DESCRIPTION</code>	Brand affiliation
<code>PRODUCT_DESCRIPTION</code>	Fuel-type label (ULP, 98 RON, Diesel, LPG, ...)
<code>PRODUCT_PRICE</code>	Posted retail price (cents per liter)
<code>ADDRESS</code>	Street address

LOCATION	Suburb / locality
POSTCODE	Postcode (4-digit)
AREA_DESCRIPTION / REGION_DESCRIPTION	Government-defined regional grouping (used as a secondary geographic check)

1.4 Costco gas station opening dates (hand-collected)

We hand-compiled a 10-row table of every Costco fuel station in Australia, recording each station's state, suburb, postcode, coordinates, and treatment date. Each treatment date is the station's first-observed date in the relevant state's price registry, then cross-validated against announced fuel-station opening dates and trade press (Section 2(d)). The composite unique key is name. Of the 10 stations, four are usable as treated units in the synthetic-control analysis (Coomera, Casuarina, Perth Airport, Lake Macquarie); the remaining six are excluded from formal estimation but appear as descriptive context.

Outcome and treatment construction

Our main outcome is the monthly mean unleaded retail price (cents per litre), constructed in Section 2 by filtering to regular unleaded (FuelCode in {U91, E10, P91} for NSW; Fuel_Type in {Unleaded 91, e10} for QLD; PRODUCT_DESCRIPTION = ULP for WA) and aggregating in two stages: (station $\times$ month), then across stations within the unit. The unit differs by group: each *treated* unit is a (Costco $\times$ month) cell averaging non-Costco stations within 5 km of one of the four treated Costco coordinates; each *donor* unit is a (postcode $\times$ month) cell averaging non-Costco stations within the postcode. The treatment indicator (a per-Costco binary post-opening flag) is constructed in Section 2 by merging the hand-collected treatment-date table onto the treated panel. The donor pool consists of postcodes located more than 20 km from any Costco that also clear the downstream coverage filters (Section 2(g)).

Summary statistics, postcode-month analysis sample (3 states, Jan 2018 – Apr 2026)

Variable	n	Mean	SD	Min	p25	Media n	p75	Max
<i>Main outcome</i>								
Mean unleaded price (¢/L)	17,953	160.5	28.4	92.0	137.3	162.2	183.6	259.0
<i>Main covariates</i>								
Station distance to nearest Costco (km)	5,980	204.8	357.4	0.0	14.8	48.3	216.7	2,214

Counts: 84 stations classified as treated (within 5 km of a treated Costco), 3,926 donor-eligible (>20 km from every Costco), 1,970 in the 5–20 km exclusion donut. The 17,953 postcode-month rows above are the donor candidate panel before geographic and quality filters; the post-filter SC donor pool is 17,578 cells (Section 2(g)).

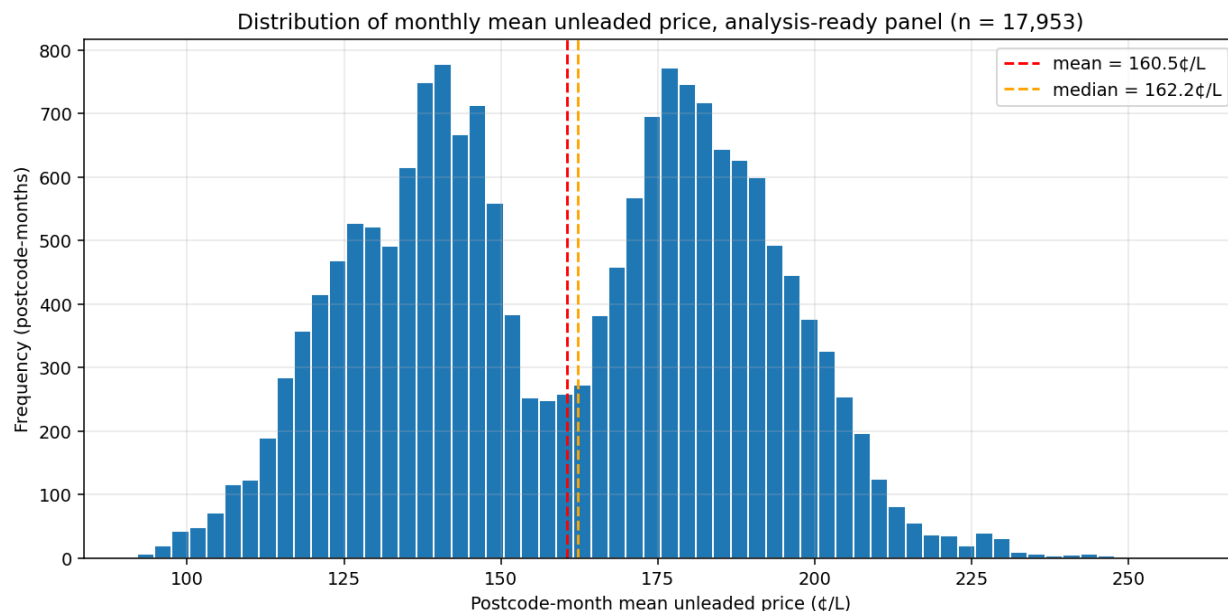


Figure 1. Distribution of postcode-month mean unleaded prices across the 17,953 panel observations. The distribution is approximately bimodal: the lower mode reflects the 2018–2020 price regime, the higher mode reflects the post-2022 elevated price era. Mean = 160.5 ¢/L, median 162.2 ¢/L. No observations outside 92–259 ¢/L after our 50–500 ¢/L sanity filter.

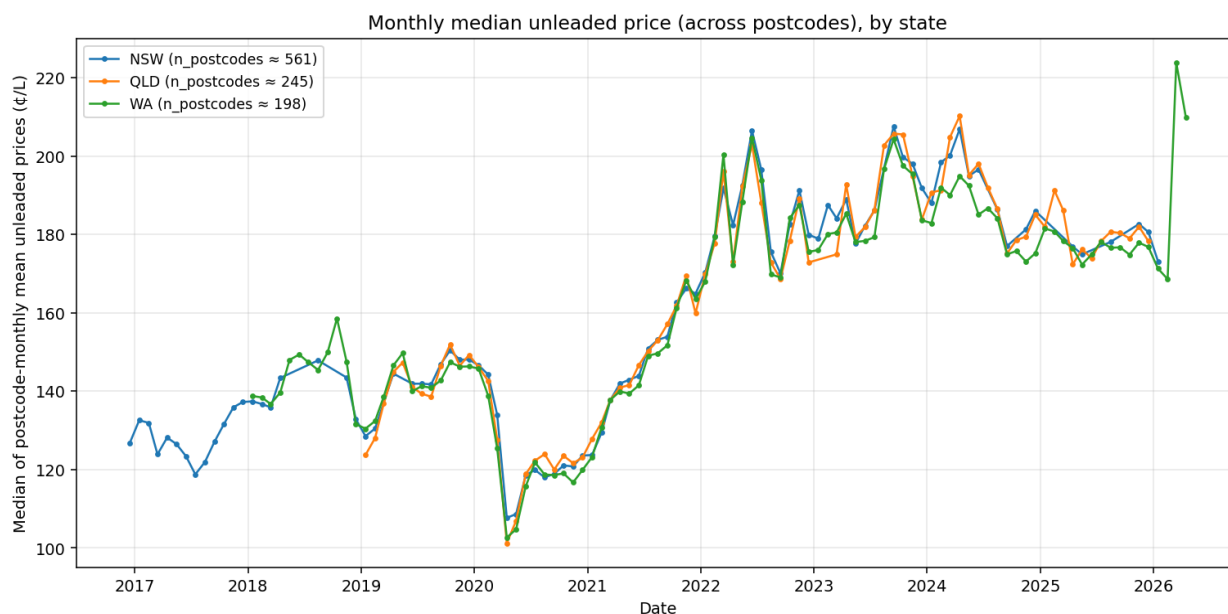


Figure 2. Median of postcode-monthly mean unleaded prices over time, by state, January 2018 to April 2026. All three states track each other closely throughout: prices fall in 2018–2020 (peaking with COVID demand collapse in early 2020), rise sharply through 2022 (Russia/Ukraine energy shock), dip during the Mar–Sep 2022 fuel-excite cut, and remain elevated through 2026. State-level co-movement supports our use of within-state donor pools to absorb common national shocks.

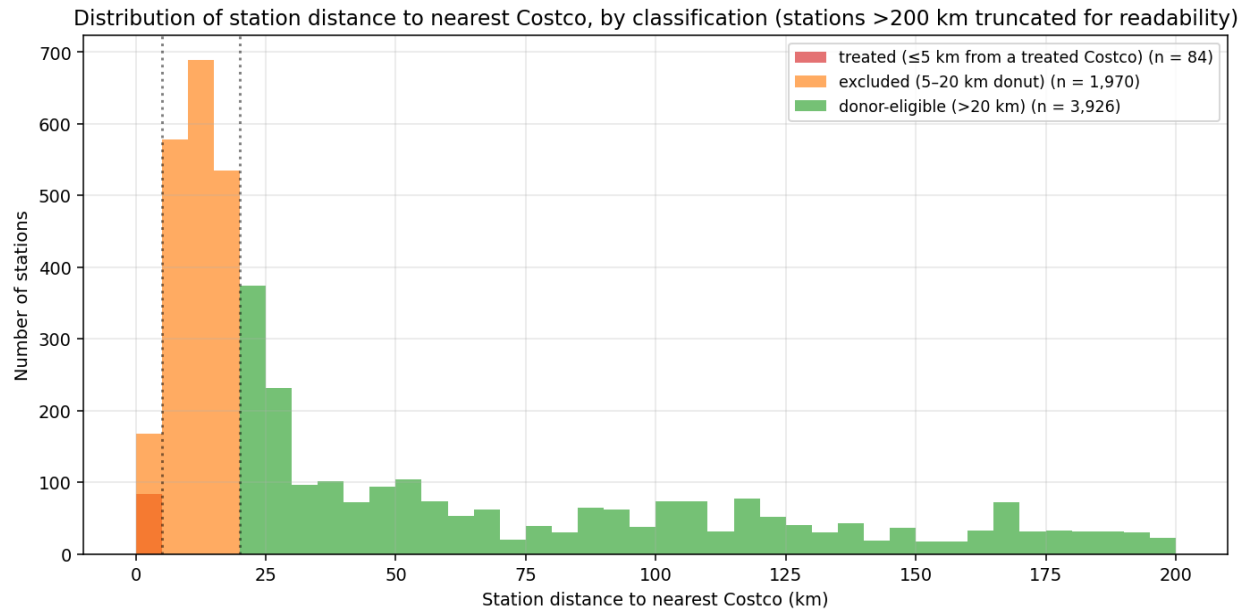


Figure 3. Distribution of station distance to the nearest Costco, split by classification. Vertical dotted lines mark the 5 km treated boundary and the 20 km donor-pool boundary; stations 5–20 km from any Costco are excluded from both groups to avoid spillover contamination. The bulk of donor-eligible stations sit well outside any plausible Costco competitive footprint, supporting the donor pool’s status as an untreated counterfactual. Stations beyond 200 km are truncated for readability.

Australian Costco fuel stations by treatment status

Costco	State	Treatment date	Pre months	Post months	Status
Coomera	QLD	May 2023	51	31	Treated
Casuarina	WA	Nov 2022	58	42	Treated
Perth Airport	WA	Feb 2020	26	74	Treated
Lake Macquarie	NSW	May 2022	58	35	Treated
Marsden Park	NSW	Jul 2017	11	105	Excluded (pre too short)
Auburn	NSW	Aug 2025	107	8	Excluded (post too short)
Canberra Airport	ACT	Nov 2023	0	29	Excluded (no pre)
Casula	NSW	Dec 2016	4	113	Excluded (pre-registry)
North Lakes	QLD	Jan 2019	1	89	Excluded (pre-registry)
Ipswich	QLD	Mar 2019	3	87	Excluded (pre-registry)

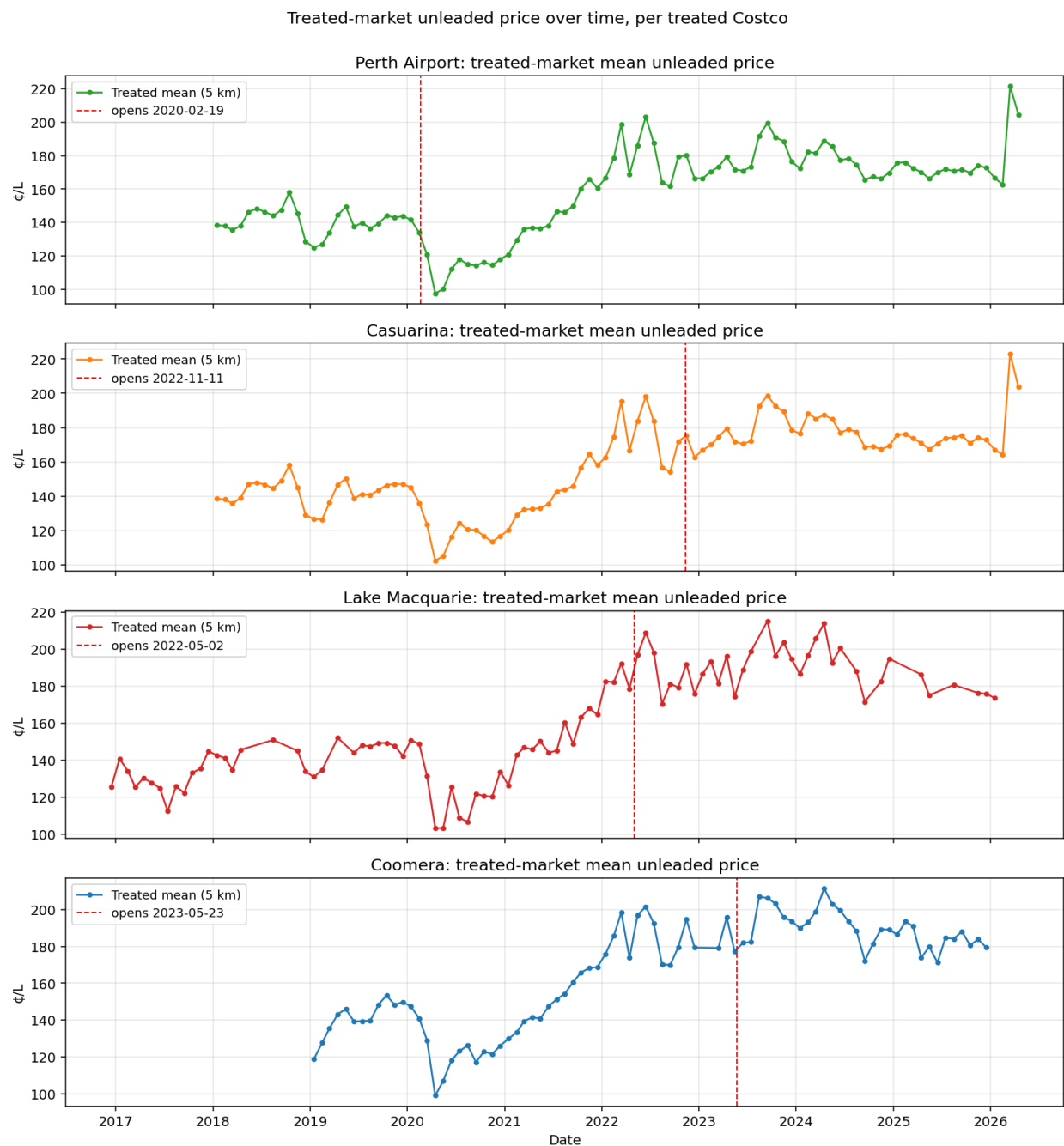


Figure 4. Treated-market mean unleaded price over time for each of the four treated Costcos, with the vertical red dashed line marking the validated treatment date (Section 2(d)). Eyeballing pre-period trajectories before fitting any model surfaces data-quality issues and high-leverage points: Casuarina’s sparse treated-station count (~3 stations) shows up as month-to-month volatility that the other three Costcos don’t exhibit; Coomera’s registry coverage gradually builds up through 2018–2020 before stabilizing.

Data quality issues

Price units differ across registries. NSW prices are in cents/L (e.g. 158.7), QLD prices are in tenths of cents (1899 = 189.9 ¢/L), WA prices are in cents/L. Each registry’s ingest applies the appropriate

scaling so the panel is uniform in cents/L.

Coordinates missing in historical NSW and WA files. NSW FuelCheck XLSX records carry only address text; WA FuelWatch records carry address and postcode (no lat/long). QLD historical records do include coordinates inline. We resolved this by pulling the live NSW FuelCheck API (/v1/fuel/prices) and the live WA FuelWatch API (/api/sites) once and joining historical records to the live coords by (normalized_name, postcode). The live-API snapshot contains NSW 3,281 stations, WA 921 stations, QLD 1,882 stations, totalling 6,084. Stations that closed, rebranded, or were re-numbered before April 2026 are absent from the snapshot, so historical rows for them fail the join: the snapshot misses 2,376 of 4,543 NSW historical stations (52.3%) and 673 of 1,423 WA historical stations (47.3%); QLD is unaffected because its historical files carry coordinates inline. The loss falls almost entirely on the donor pool: all four treated Costcos join cleanly. Row-level impact on the analysis panel is in Section 2(g).

Brand renaming in NSW (May 2022). Every NSW Costco station was renamed in the registry from e.g. "Costco Marsden Park" to "Costco Marsden Park (Members only)" on 2 May 2022. We strip the "(Members only)" suffix before matching so the rename is invisible to the analysis pipeline.

Outliers in price. Quotes below 80 ¢/L or above 280 ¢/L are implausible for Australian retail unleaded post-2018 (national averages hover 130–220 ¢/L). After our 50–500 ¢/L sanity filter, the post-aggregation panel of 17,953 postcode-months has no observations outside 92–259 ¢/L, well within plausible Australian range.

Treated-Costco coverage variance. Coomera's pre-2020 QLD coverage is sparse because the QLD scheme began in December 2018 and Coomera-area stations entered the registry gradually; the relevant pre-period (Sep 2020 → May 2023) is fully covered. Casuarina has only ~3 unique stations within 5 km contributing to the treated mean, reflecting a sparse coastal residential market; coverage is 100 % but the treated mean is noisier than for the other three Costcos. Lake Macquarie has a small number of intermittent monthly gaps (largest 3 months) that do not impair pre-period fitting.

Donor pool construction. Out of all postcodes with any non-Costco unleaded data, the donor pool drops 362 postcodes within 20 km of any Costco (potential contamination from Costco's competitive footprint), 446 with fewer than 3 stations on average per month (means too noisy), and 0 with fewer than 24 months of observations. **196 donor postcodes** survive all filters: NSW 104, QLD 56, WA 36, totalling 17,578 (postcode × month) panel observations. Per-state breakdown of the 362 / 446 / 0 drops is in Section 2(g).

Multiple records per station-day. NSW and QLD both record price-update events rather than daily snapshots, so a station appears multiple times within a day if it changed prices on multiple fuel types. These are not duplicates; they are legitimate event-level records and are correctly aggregated to monthly means by the panel build.

Contingencies

If histograms of postcode-month unleaded prices reveal extreme outliers in any donor postcode (e.g. a one-time data-entry anomaly from a single station dominating a small postcode), the panel is winsorised at the 1st and 99th percentile before synthetic-control weight fitting. Postcode-months with fewer than 3 contributing observations after the panel build are flagged as missing, not retained at face value. If pre-period fit quality for any treated Costco is demonstrably poor (Section 3, visible mismatch between

treated and synthetic in the pre-period), that Costco is presented with a documented limitation, not as a headline result; Casuarina is the most likely candidate given its small treated-station count.